

Israel Souza Ferreira

Machine Learning Researcher | ML Engineer

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PORTFOLIO

isrreal.github.io — projects, technical documentation, demos, and full resume.

PROFESSIONAL EXPERIENCE

Tieta Artificial Intelligence

R&D Researcher, AI for Proficiency Analysis — CNPq RHAe Research Fellow

Remote, Brazil
Apr 2025 – Present

- Modeled characteristics of educational questions with **Item Response Theory**, validating model-estimated signals against reference psychometric parameters.
- Built inference pipelines with **language models** orchestrated by **DSPy**, including procedural dialogue generation and cognitive-trajectory tracking.
- Compared alternative approaches through paired evaluation on the same items, measuring both the correlation of the resulting signals with reference parameters and the inference cost of each.
- Experimented with relative representations and attention-based reducers in **multitask regression** architectures over textual and visual data.
- Structured reproducible experiments, evaluations, and technical documentation to support the research cycle.

Contract Project — Educational Institution

Machine Learning and Backend Developer

Remote, Brazil
2026 – Present

- Independently delivered a contracted end-to-end facial recognition attendance system, functional and in final acceptance testing before deployment, covering machine learning, backend development, databases, security, testing, and deployment.
- Implemented 1:1 verification and 1:N facial identification using InsightFace, embeddings in $\mathbb{R}^{512}$, pgvector, multi-frame aggregation, liveness detection, and anti-replay protection.
- Trained an EfficientNet-B0 classifier for medical-document triage using Optuna, stratified validation, AMP, and a hybrid CNN + OCR decision pipeline; diagnosed underfitting through controlled ablation and raised validation accuracy by **13 percentage points**.
- Built a FastAPI backend with JWT authentication, geofencing, PostgreSQL, SQLAlchemy, and Alembic; delivered Dockerized infrastructure with HTTPS, GitHub Actions, and **142 automated tests**.

Federal University of Ceará — Quixadá Campus

Teaching Assistant for Algorithms, Calculus, and Pre-Calculus

On-site, Brazil
2022 – 2024

- Provided academic support in data structures, algorithms, complexity analysis, and calculus applied to Computer Science.

SELECTED PROJECTS

- Triple Roman Domination in Graphs:** first metaheuristic approaches reported in the literature for the problem, using a Genetic Algorithm, MAX-MIN Ant System, RVNS, and Integer Linear Programming for experimental validation.
- Dengue Forecasting System:** MLOps pipeline under development for dengue case forecasting using Brazilian SINAN surveillance data, LSTM, PLE, PostgreSQL, MLflow, FastAPI, and Docker; reduced RMSE from approximately 98 to 83 for the Ceará time series.
- Brazilian Emergency Aid — Query Benchmark:** system under redesign for benchmarking SQL indexing strategies over **257,170,290 records totaling 31.6 GB**, evaluating B-Tree and GIN/TRGM indexes across searches, aggregations, and joins using PostgreSQL, FastAPI, Streamlit, and Docker.

TECHNICAL SKILLS

Languages: Python, C++, SQL, and Bash.

Machine Learning: PyTorch, Scikit-learn, DSPy, FAISS, Optuna, InsightFace, OpenCV, Pandas, NumPy, and SciPy.

Backend, Data, and MLOps: FastAPI, PostgreSQL, pgvector, SQLAlchemy, Alembic, Streamlit, Docker, Git, GitHub Actions, GitLab CI/CD, MLflow, pytest, and mypy.

Methods: multimodal and multitask learning, MoE/MMoE, LSTM, attention, vector search, Item Response Theory, B-Tree and GIN/TRGM indexing, hypothesis testing, and ablation analysis.

EDUCATION

Federal University of Ceará — Quixadá Campus

B.Sc. in Computer Science

Brazil
Expected Jul 2026

LANGUAGES

Portuguese: Native | **English:** Fluent technical reading of research papers and documentation